

Clinical Risk Assessment Report: Patient 23

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 8.9% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.40x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.0%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -1.99 [Bottom 4%] (-2.0%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Patient Age (Age) **Value: 68.0 [Avg (68th %ile)] (-1.2%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.5 [Avg (33th %ile)] (-0.9%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.12 [Avg (64th %ile)] (-0.9%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Carcinoembryonic Antigen (CEA) **Value: 0.7 [Bottom 1%] (-0.9%)**

Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.09 [Avg (41th %ile)] (-0.8%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.01 [Avg (38th %ile)] (-0.7%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.45 [Avg (30th %ile)] (-0.6%)**

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Machine Learning Feature Attribution (SHAP Decision Path)

